FABIAN MOA

CFA® Program

Level III

PORTFOLIO MANAGEMENT PATHWAY

FORMULA SHEET (2025) Version 1.3

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FOR REFERENCE ONLY

(Note: Formula Sheet is not provided in the CFA exam)

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CFA Level III – Formula Sheet (2025)

Setting Up the Texas BA II Plus Financial Calculator

Video: https://youtu.be/0MS8d8QOFmc

VOLUME 1 – ASSET ALLOCATION

Learning Module 1: Capital Market Expectations, Part 1: Framework and Macro Considerations

Real %ΔGDP

$$\text{Aggregate trend growth of economy} = \text{Growth from labor inputs} + \text{Growth from labor productivity}$$

where:

$$\text{Growth from labor inputs} = \text{Growth in potential labor force size} + \text{Growth in actual labor force participation}$$

$$\text{Growth from labor productivity} = \text{Growth from increasing capital inputs} + \text{Growth in total factor productivity}$$

Growth in Real GDP = Growth from Labor inputs + Growth from Labor Productivity → may have to derive by re-arranging.

Aggregate market value of equity

$$V_t^e = GDP_t \times S_t^k \times PE_t$$

$$\text{Equity return} = \text{Capital appreciation} + \text{Dividend yield} \quad \left(\equiv \frac{V_{t+1}^e}{V_t^e}\right)$$

Over a finite horizon:

$$\text{Equity return} = \%\Delta GDP_t + \%\Delta S_t^k + \%\Delta PE_t + \text{Dividend yield}$$

→ on a per year basis; → all these on forecast basis

In the long-run:

$$\text{Equity return} = \%\Delta \text{Nominal } GDP_t + \text{Dividend yield}$$

OR, = Growth in Eq M-cap + Div. Yld

where:

V_t^e = Aggregate market value of equity

GDP_t = Nominal level of GDP = real GDP + (Forecasted inflation)

S_t^k = Share of profits in the economy (earnings) = $\frac{EPS_t}{GDP_t}$ → may need to incorporate growth to get future aggr V_t^e.

PE_t = P/E ratio

$$\text{Dividend yield} = \frac{\text{Dividend payout ratio}}{\text{Profit multiple}}$$

→ like P/E ratio

$$\frac{div}{EPS} \times \frac{EPS}{P_x}$$

⇒ Higher earnings Mult will produce lower Div Yld.

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Cannot simult. have fixed exchg rate, unrestricted cap. flows, & pursue independent monetary policy.

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Taylor Rule (Video: https://youtu.be/CI_SShKmOwA)

Nominal: $i^* = r_{neutral} + \pi_e + 0.5(\rho_e - \rho_{trend}) + 0.5(\pi_e - \pi_{target})$ — output Gap; Inflation Gap

where:

i^* = target nominal policy rate

$r_{neutral}$ = real neutral policy rate

π_e = expected inflation rate

π_{target} = target inflation rate

ρ_e = expected real GDP growth rates

ρ_{trend} = trend real GDP growth rates

$i^* - \pi_e$ = real, inflation-adjusted policy rate

$$X - M = (S - I) + (T - G)$$

where:

$X - M$ = Net exports

S = Savings

I = Investment

$T - G$ = Government surplus = taxes − govt spending; If $T-G<0 \Rightarrow$ fiscal deficit. If deficit ↑ ⇒ Loose

Learning Module 2: Capital Market Expectations, Part 2: Forecasting Asset Class Returns

Grinold-Kroner model, $E(R_E) = \left(\frac{D_1}{P_0} - \%\Delta S\right) + \%\Delta E + \%\Delta P/E$ = Income Ret^n + Nom. earnings growth + Repricing Ret^n

Expected equity return (eg. over 10 yrs)

$$E(R_e) \approx \frac{D}{P} + (\%\Delta E - \%\Delta S) + \%\Delta P/E$$

where:

$\frac{D}{P}$ = Dividend yield — ↑p. cap. gain.

$\%\Delta E$ = Expected % change in total earnings = nominal GDP growth + %Δ(E/GDP) → if multi year growth → $\left(\frac{y_2}{y_1}\right)^{1/n} - 1$

= Expected nominal earnings growth return

$\%\Delta S$ = Expected % change in shares outstanding ($\%\Delta S < 0$: Net share repurchases) → dec in shares o/s

$\%\Delta P/E$ = Expected % change in price-to-earnings ratio; reported as P/E expansion over fk; use p.a. chng.

= Expected repricing return = Rate of P/E ratio expansion.

$\%\Delta E - \%\Delta S$ = Growth rate of earnings per share → NOTE ∵ $EPS = \frac{E}{S}$

$\frac{D}{P} - \%\Delta S$ = Expected cash flow ("income") return

$\%\Delta E + \%\Delta P/E$ = Expected capital gains

In the long run, $\%\Delta E$ = Nominal GDP growth, $\%\Delta S = 0$, $\%\Delta P/E = 0$ OR: $\%\Delta E = \left\{\text{Nom. GDP growth} + \%\Delta\left(\frac{E}{GDP}\right)\right\}$

Video: https://youtu.be/yOmaMz2YC18 — ∴ equity ret^n = ΔGDP% + Div Yld. ②③

(+) Impact of inc div. payout ratio

Fisher Effect: ΔS = infl exp (D − F); long-term − v. slow.

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Risk Premium Approaches to Equity Returns

Equity-vs-bills premium = Term premium + Equity-vs-bond premium

↳ Term premium = Return on bonds – Return on bills

OR, Tot Ret^n = curr. div yld + Exp. earnings growth rate + (Hist. equity risk prem − exp. inflation)

Singer-Terhaar Approach ↳ assumes markets are perfectly liq; adj with illiq prem esp for RE.

Risk premium on asset class i under full integration:

$$RP_i^G = \rho_{i,GM}\sigma_i\left(\frac{RP_{GM}}{\sigma_{GM}}\right) + Illiquidity\ premium$$ — (a) eg real estate.

($\because \beta = \rho_{i,m}\frac{\sigma_i}{\sigma_m}$) → @19

Risk premium under complete segmentation: → same as (a) but $\rho = 1$ & Sharpe of segmted mkt. GM

$$RP_i^S = 1 \times \sigma_i \times \frac{Sharpe\ ratio\ of}{segmented\ market} + Illiquidity\ premium$$ — (b)

Singer-Terhaar risk premium on asset class i: $RP_i = \phi RP_i^G + (1-\phi)RP_i^S$ → (c) weighted of (a) & (b)

Expected return of asset class i, $E(R_i) = R_F + RP_i$ → recon. to add R_F to RP to get exp. ret^n

where:

$\rho_{i,GM}$ = correlation between *i*th asset and global market portfolio

σ_i = standard deviation of *i*th asset's return — eg. equity class.

$\frac{RP_{GM}}{\sigma_{GM}}$ = Sharpe ratio of global market portfolio

ϕ = Degree of integration

R_F = Risk-free rate

Note:

- If Sharpe ratio of segmented market not given, use Sharpe ratio of global market portfolio → maybe provided as SR_i

Video: https://youtu.be/RK2WETqIzoQ

Conservatism – slow to update prior views when presented w. new info. Representativeness: ov'wt navi info based on past exprnc, model, heuristic & Assume small samples r representative of populations. → Return chasing.
Loss Aversion: people prefer avoiding losses v/s achieving gains. Utility from gain is << utility given up w. equivalent loss.
Illusion of Ctrl: cognitive bias for the tendency to overestimate one's ability to control events.

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TIPS = exp. protected secu that pay fixed real rate on an inc princ. If exp infl < actual infl, TIPS will slightly outperform fix r bnd which will outperform fixed rate bonds. & Vice-versa if actual infl < exp. infla. Nom fixed rate bonds, protect agnst expected infla. But infla risk deals with unexp. infla that exceeds exp. infla compo of nom yld.

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Risk Premium (Building Block) Approach to Fixed Income Returns

Higher real trend grwth in GDP ⇒ higher nom avg yld

Bond	Required Return
Short-term fixed-rate government bill	Real risk-free rate + Inflation premium
Long-term fixed-rate government bond	Real risk-free rate + Inflation premium + Term premium + Liquidity premium
Long-term inflation-linked government bond	Real risk-free rate + Term premium + Liquidity premium
Long-term fixed-rate corporate bond	Real risk-free rate + Inflation premium + Term premium + Liquidity premium + Credit premium
Long-term callable fixed-rate corporate bond	Real risk-free rate + Inflation premium + Term premium + Liquidity premium + Credit premium + Call risk premium

remove inflan prem: eg. TIPS

inflation is accounted for in nominal yield; not required

Appraisal-based ret data: No bias in mean ret; vol. is underestimated.

Real Estate REIT - direct ownership of properties; like eq. in short term; like direct RE in LT over multi-period.

Adjusting for smoothing ine corr btw REIT & direct RE.

Long-run:

$$E(R_{re}) = Cap\ rate + NOI_{\text{Nominal}}\ growth\ rate$$

Alter: Swinger-Tehran model @ 5.
→ nominal terms = real + inflan

Cap Rate: reflect LT disc rate; procyclical: rate ↑ when eco expansion; ↑ rely corr with cred spred; Expan. eco ⇒ lower liq prem on RE ⇒ lower cap rate

Finite horizon: Cap Rates r pro-cyc with int rates → rise & fall with avg yields. Higher availbty of debt/GDP ⇒ lower cap rt

$$E(R_{re}) = Cap\ rate + NOI_{\text{Nominal}}\ growth\ rate - \%\Delta Cap\ rate$$

$$\text{Cap Rate} = \frac{NOI}{\text{Prop. Value}}$$

Capital Flows

In the **long run** (diff in nominal ST rates)

$$E(\%\Delta S_{d/f}) = (r^d - r^f) + (Term^d - Term^f) + (Credit^d - Credit^f) + (Equity^d - Equity^f) + (Liquid^d - Liquid^f)$$

(Equity: ERP; diff in risk prem btw dom & foreign p/fo. [T, C, E, L])

Rem

Currency

For a currency pair, d/f, if f changes by x% against d, then d changes by $\frac{1}{1+x} - 1$ against f.

UncovIRP (UIRP): $E(\%\Delta S_{d/f}) = r^d - r^f$ // Nominal

VCV Matrix with Sample Statistics

With N assets, required:

- N variances
- $\frac{N(N-1)}{2}$ covariances

$d_1 = (1+x)f$

$$\frac{d\left(\frac{1}{1+x} - 1\right)}{f(1+x)} = \frac{-x\,d}{1+x}\,f$$

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VCV Matrices from Multi-Factor Models

Return on *i*th asset:

$$r_i = \alpha_i + \sum_{k=1}^{K} \beta_{iK} F_K + \varepsilon_i$$

Variance of the *i*th asset:

$$\sigma_i^2 = \sum_{m=1}^{K} \sum_{n=1}^{K} \beta_{im}\beta_{in}\rho_{mn} + \nu_i^2$$

Covariance between *i*th and *j*th asset:

$$\sigma_{ij} = \sum_{m=1}^{K} \sum_{n=1}^{K} \beta_{im}\beta_{jn}\rho_{mn}$$

→ excludes the idiosyncratic term of the above & changes to i,j instead of i & i.

With N assets and K factors, required:

- $(N \times K)$ factor sensitivities
- $\frac{K(K+1)}{2}$ factor covariances

where:

α_i = Intercept
β_{iK} = Asset's sensitivity to the *k*th factor
F_K = *k*th common factor return
ε_i = stochastic term (mean = zero)
ρ_{mn} = correlation between *m*th and *n*th factors
ν_i^2 = variance of unique component of *i*th asset's return

Video: https://youtu.be/XVpJ8yuTnqo

Smoothed Returns

$$R_t = (1-\lambda)r_t + \lambda R_{t-1}$$

also in (13)

$0 < \lambda < 1$ = serial correlat^n

$$var(r) = \left(\frac{1+\lambda}{1-\lambda}\right) var(R)$$

where:

R_t = Current observed return
r_t = Current true return
R_{t-1} = Previous observed return
$var(r)$ = True variance as a func of var on smoothed ret^n

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ARCH Model

$$\sigma_t^2 = \gamma + \alpha\sigma_{t-1}^2 + \beta\eta_t^2$$
$$= \gamma + (\alpha + \beta)\sigma_{t-1}^2 + \beta(\eta_t^2 - \sigma_{t-1}^2)$$

$\gamma, \alpha, \beta \geq 0$, $\alpha + \beta < 1$ (shocks to vol arise from unexpectedly large or small returns.)

where:

σ_t^2 = Variance in period t

η_t = Unexpected component of return in period t (mean = 0)

$\eta_t^2 - \sigma_{t-1}^2$ = Shock to variance in period t

Unconditional expected value of variance

$$\frac{\gamma}{1 - \alpha - \beta}$$

Capital Ⓐ: generate income on an ongoing basis → div, int payst.
Transformable Ⓐ; / consumable Ⓐ: transformed while producing value
Store of Val Ⓐ: realise val thr sale or exchge ; eg. stamps

Reverse Optim (RO): estimate exp. ret for forecasting → ∴ lower estim err from MVO; RO retains all E(R) estim to a common risk fctr → systematic risk. Those E(R) used in MVO → well-behv MA. R-MVO produces E(R) estim that are then used in an MVO. Rev MVO uses m-cap wtg; only 1 mkt risk prem (MRP) used for all ast clss; use subjective E(R) in BL first, then use E(R) that BL outputs as input to MVO.

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Learning Module 3: Overview of Asset Allocation

Asset only approach: ∅ liab; focus on (A) side of BS + PV of human cap as a pfo of (A). Rem MVO. Typically obj Max Sharpe @ given level of risk. *Economic Net Worth = Economic Assets − Economic Liabilities* → see also (L)B

$$\begin{cases} \textit{Economic Assets} = \textit{Financial Assets} + \textit{Extended Assets} \\ \textit{Economic Liabilities} = \textit{Financial Liabilities} + \textit{Extended Liabilities} \end{cases}$$

Liab. Relative Appr: explicitly consider liab – legal & quasi (eg. PV of retirmt expn). eg. Surplus optimizers.

Learning Module 4: Principles of Asset Allocation

Both R-MVO & BL produce E(R) estim → used in MVO.

Mean-Variance Optimization (MVO): uses single-period framwk; can be sensitive to small chngs to input pmtr. MVO produces less diversified pfo than Black-Litterman (BL). BL incorp greater consistency btw asst's exp. ret & its contri to systematic risk. Unlike BL, MVO does NOT allow investors to incorp their unique ret exp for (A) cl

Utility

$$U_m = E(R_m) - 0.005\lambda\sigma_m^2$$

BL model can be used with/wo any (non-negativity) constraints.

where:

U_m = Investor's utility for asset mix, m

R_m = Return for asset mix, m ← Normal distbn → ignore the presence of (L); uses S.Dev for risk → ID now; poor.

λ = Investor's risk aversion coefficient → MVO asst allocn tend to concen. among few (A)cl even if many diversn

σ_m^2 = Expected variance of return for asset mix m (in %) MVO uses PSV pfo vol. as primary measure of risk. BL starts w. excess retn from R.D. & provides technique to incorp investor's own views. ∴ get well diversified AA from such exp. ret.

✓ *Surplus = Market value of assets − Present value of liabilities*; Proj Benefit Oblig (PBO) = uses proj pay & includes benefits earned due to tenure of service. Track plan's funded status v/s m.v. of (A).

$$\textit{Funding ratio} = \frac{\textit{Market value of assets}}{\textit{Present value of liabilities}}$$

Accumulated Benefit Oblig (ABO) = sponsor's current legal oblign to participants if plan were to close or convert to DC.

Surplus Optimization

$$U_m^{LR} = E(R_{s,m}) - 0.005\lambda\sigma_{s,m}^2$$

where:

U_m^{LR} = Surplus objective function's expected value for asset mix m; applies a linear corr. assmpn.

$E(R_{s,m})$ = Expected surplus return for asset mix m

$$= \frac{\textit{Change in asset value} - \textit{Change in liability value}}{\textit{Initial asset value}}$$

Note: ∴ SO extends MVO to include (L) under single period. But: Integrated ALM allows analysis of multiple period. Does not req linear corr. Assmpn

liab-relative measure of risk. $\sigma_{s,m}$ = Surplus return volatility for asset mix m

λ = Investor's risk aversion → Integ ALM applicable to any funded retr, can implemt via factor model, link (A)(L) to obj. factors.

Cash in MVO: a) asset cl w v. short horizon & min. volatility (~MMF)
b) Two Fund Separn: optimize risky asset excluding cash, det highest SR by combining risky pfo w/ risk-free cash. Adding cash or borrg won't change Sharpe Ratio.

Goals-based Asset Allocation — reqd inputs = disc rate, time horizon, reqd prob. of success. → based on want, need v/s desire → reflects risk aversion

Video: https://youtu.be/ufo0cNWmfbo

Safety-First (SF) Ratio = $\frac{E(R) - R_L}{\sigma}$; R_L = Min acceptable retn; E(R) = exp. retn; σ = Std. dev. of retn.

Resampled Efficient Frontier Approach → 2 advtgs over MVO = 1) optimal pfo more diversified; 2) Pfo wts more stable over time

Mean CVaR option: incorp d/side risk; Min pfo CVaR; With attr → left-tailed risk relevant.

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Hedging/Ret" Seeking Pfo.: used by conservative investors; highest prob of closely matching (A) vs (L); Min downside risk. - only for overfunded plans; U/funded plan does not have leeway to fully hedge (L), & invest surplus in risk portion.

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Risk Parity – each (A) class shld make an equal contr to total risk.

$$w_i \times Cov(r_i, r_p) = \frac{1}{n}\sigma_p^2$$

→ RHS is fixed → asset with greater than avg cov. w. pfo must have a lower wt to offset the higher contribut" to tot risk.

where:

w_i = Weight of asset i

$Cov(r_i, r_p)$ = Covariance of asset i with portfolio ✓

n = Number of assets

σ_p^2 = Variance of portfolio

Risk Budgeting AA ~~from~~ is optimal from risk budgeting persp. when $\left[\frac{E(R) - R_f}{MCTR}\right]$ is same for all asset classes & is equal to Sharpe of pfo.

Marginal Contribution to Risk

$$MCTR = \underset{\text{to portfolio}}{\text{Asset beta relative}} \times \underset{\text{deviation}}{\text{Portfolio standard}} = \beta_i \times \sigma_p$$

Actual Contribution to Risk

$$ACTR = \frac{\text{Asset weight}}{\text{in portfolio}} \times MCTR = w_i \beta_i \sigma_p$$

$$\text{Ratio of excess return to MCTR} = \frac{\text{Expected return} - \text{Risk free rate}}{MCTR}$$

→ set to same for all (A) cl. in pfo. & equal to Sharpe ratio of pfo.

Learning Module 5: Asset Allocation with Real-World Constraints

Tighter bands for rebalancing needed if: low trn cost, low corr w. pfo, mean reverting, located in tax-deferred ac

After-tax Portfolio Optimization

Time-based rebal: defined wit rebal. intervals; not sensit to mrkt.
Threshold " " : sophist monitorg + auto^
Proportional % Rebal: rebal range ∝ wt. of asset cl.

$$r_{at} = r_{pt}(1 - t)$$

$$\sigma_{at} = \sigma_{pt}(1 - t)$$ ✓

$$RR_{at} = \frac{RR_{pt}}{1 - t}$$

→ inverse r'ship, unlike ret" & risk [r, σ]

where: r_{at} = Expected after-tax return
r_{pt} = Expected pre-tax return
t = Expected tax rate
σ_{at} = Expected after-tax standard deviation
σ_{pt} = Expected pre-tax standard deviation
RR_{at} = After-tax rebalancing range ✓
RR_{pt} = Pre-tax rebalancing range

TAA is based on how ~~blt~~ ret" on (A) classes deviate from LIT CME for SAA. Allow deviat" from policy wts within a range.
DAA (dynamic) = deviation from SAA due to LT volatn or eco signals.

If taxable investor ⇒ turnover shld be low → passive ; If tax-exempt ⇒ t/o can be hi ⇒ Active Mgmt.

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VOLUME 2 – PORTFOLIO CONSTRUCTION

Well constructed pfe: risk expo aligned with investor expectn & constr; less idio (unsys) risk rel. to Tot Risk. If 2 pfolios have comparable factor expos, pfolio with lower abs vol. & lower activ risk will be prefrd. If 2 pfolios have similar activ & abs risk, similar costs, & alpha skills are similar, the pfolio with higher activ share is prefrd. It leverages a skill of Mngr & will have higher exp. returns.

Learning Module 1: Overview of Equity Portfolio Management

Herfindahl Hirschman Index (HHI) → maybe in an index too.

$$HHI = \sum_{i=1}^{n} w_i^2$$

where: w_i = Weight of stock i in portfolio

$$\text{Effective number of stocks} = \frac{1}{HHI}$$

$$\text{Tracking error}_p = \sqrt{\text{Variance}_{R_p - R_b}} = \sigma_{R_p - R_b}$$

Arith $\text{Excess return}_p = R_p - R_b$ → see also ⑰ for comm Exc Ret

Learning Module 2: Overview of Fixed Income Portfolio Management

for Par Bond: coupon == YTM

$V_{callable} = V_{st} - call\ op$; $V_{putt} = V_{st} + Put\ Opn.$

Expected fixed-income return

$$E(R) = \text{Yield income} + \text{Rolldown return}$$
$$+E(\text{Change in price based on investor's view of benchmark yield})$$
$$+E(\text{Change in price based on investor's view of yield spreads})$$
$$+E(\text{Change in price based on investor's view of currency value changes})$$

ΔP_x (b/mk) ΔP_x (y/s) → # ΔP_x (Fcy)

$$\text{Yield income (or current yield)} = \frac{\text{Annual coupon payment}}{\text{Current bond price}}$$ like Div Yld

$$\text{Rolldown return} = \frac{\text{Price}_{End} - \text{Price}_0}{\text{Price}_0}$$ (Assume no change in yield curve)

N = mats x freq
I/Y = gtn/freq
PMT = FV x coupon/freq
CPT PV ?

Rolling yield = Yield income + Rolldown return

$$E\begin{pmatrix}\text{Change in price based} \\ \text{on investor's view of} \\ \text{benchmark yield}\end{pmatrix} = -ModDur \times \Delta Yield + \frac{1}{2} \times Convexity \times (\Delta Yield)^2$$

→ DR, can redo CPT PV using new vol of ytm for I/Y in calc.

Vol of ret'n on inflation-indexed bonds ↓ vol of real int rates, viz less volatile than nominal int. rates.

If a bond is more positively convex than other identical, the bond Pr will dec less than the dur'n estimate wld suggest if rate↑

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(4)

$$E\left(\begin{array}{c}\text{Change in price based} \\ \text{on investor's view of} \\ \text{yield spreads}\end{array}\right) = -SpreadDur \times \Delta Spread + \frac{1}{2} \times SpreadConvexity \times (\Delta Spread)^2$$

For foreign fixed income investments:

$$R_{DC} = (1 + R_{FC})(1 + R_{FX}) - 1, \text{ where:}$$

R_{FC} = Yield income + Rolldown return
+E(Change in price based on investor's view of benchmark yield)
+E(Change in price based on investor's view of yield spreads)

R_{FX} = Percentage change in exchange rate (DC/FC)

$$R_{FX} = \frac{E(S_{DC/FC}) - S_{0,DC/FC}}{S_{0,DC/FC}} \quad \text{(unhedged)}$$

$$R_{FX} = \frac{F_{DC/FC} - S_{0,DC/FC}}{S_{0,DC/FC}} \quad \text{(hedged)}$$

Using Leverage in the Bond Portfolio

Leveraged portfolio return

$$r_P = r_I + \frac{V_B}{V_E}(r_I - r_B)$$

OR

$$r_P = \frac{r_I(V_E + V_B) - V_B r_B}{V_E}$$

where:
r_I = return on invested funds
r_B = cost of borrowing
V_B = borrowed funds
V_E = value of portfolio's equity

Video: https://youtu.be/NadocNKzDBw

Futures

$$Leverage_{Futures} = \frac{Notional\ value - Margin}{Margin}$$

Repo

$$Dollar\ interest = Principal \times Repo\ rate \times \frac{Term\ of\ repo\ in\ days}{360}$$

Securities Lending

$$Rebate\ rate = Collateral\ earnings\ rate - Security\ lending\ rate$$

Manager of fund of one has greater auth for hvm decision (vs investor) vs SMA. In both, we expect from Mngr:

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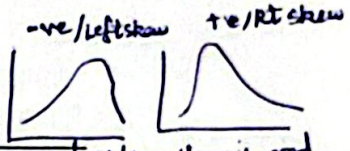
Cash Flow Matching

$$\text{Required par value for the longest maturity bond} = \frac{\text{Amount of final liability due}}{1 + \text{Coupon rate of the longest maturity bond}}$$

Video: https://youtu.be/fB257bEp59c

(32) $$\text{Convexity} = \frac{MacDur^2 + MacDur + Dispersion}{(1 + \text{Cash flow yield})^2}$$



Rt/+ve skew is good.
Left/-ve skew ⇒ few infield ⇒ Low

Learning Module 3: Asset Allocation to Alternative Investments

PvtRE plays cap. growth role but pvt credit does not; PvtRE offers hedge agnst inflation, unlike Pvt Cred.

Unsmoothed Returns

$$r_{t,unsmoothed} = \frac{r_{t,reported} - s \times r_{t-1,reported}}{1 - s}$$

rearrange from Pg 7

where: s = Serial correlation; if s is low ⇒ diff btw in vol. btw smoothed vs unsmoothed ret^n will be low. ∴ Fund with lowest smoothed vol ⇒ also the lowest unsmoothed vol.

(C) Net Asset Value (NAV)

$$NAV_t = NAV_{t-1} \times (1 + G) + C_t - D_t$$

where:

cap. commitment → paid-in capital

$C_t = RC_t \times (CC - PIC_t)$

distb^n, → $D_t = RD_t \times [NAV_{t-1} \times (1 + G)]$

Tot Ret^n = Roll ret^n + Spot Ret^n + collat ret^n = $(\Delta fut - \Delta spot) + \Delta spot$ + collat ret^n = $\Delta fut\ P_{fx}$ + Collat ret^n

Learning Module 4: Overview of Private Wealth Management

Cost-of-carry: Fut P_{fx} $F = S_e^{(r+c-y)(T-t)}$; c = cost of storage; y = convenience yld.

Incom from exp. & human capital ends at perma. retirement.

$$P_{HC} = \text{Human Capital} = \sum_{t=1}^{N} \frac{p(s_t) \times w_{t-1} \times (1 + g_t)}{(1 + r_f + y)^t}$$

= PV of all future labr earnings adj for wage growth, survival prob, incom vol.

Wt of human cap in asset Mix = P_{HC} / Tot Net wealth; Tot Net Wealth = P_{HC} + Prop Val − Mtg + Fin(A). Numr ↓ Denom

where:

$p(s_t)$ = Probability of surviving to year t

w_{t-1} = Income from employment in period $t - 1$

g_t = Annual wage growth rate

r_f = Nominal risk-free rate

y = Risk premium associated with occupational income volatility

N = Length of working life in years

Investable NW = liq assets (exclu prim resi with large mtg); Economic NW = incl human capital & intangibles like vested pension benefit. ∴ Broader than Tradit'l NW.

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Accrual equivlt aft-tax ret° is the tax-free ret°, if accrued annually & produces same AT accum° as taxable pfo's
$r(1-T_{AE}) = R_{AE}$ → get T_{AE} = accrual equiv tax rate - hypoth tax rate.

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comparing b/w taxable & tax-exempt a/c: $PV(1+R_{AE})^T = FV_{cg}$

Taxes on Income

$$Average\ tax\ rate = \frac{Tax\ liability}{Taxable\ income}$$

Accrual Taxes on Investment Returns

After-tax future value of a $1 investment, with returns taxed annually:

$$FV_{AT} = [1 + R(1 - t_x)]^T$$

where:
PV = Initial investment → used if bkd FV for PV is needed
R = Pretax return
t_x = Tax rate on investment income
T = Investment period (in years)

$$Tax\ drag = \frac{Difference\ in\ investment\ gains\ with\ and\ without\ taxes}{Investment\ gains\ without\ taxes} = \frac{(1+R)^T - [1+R(1-t_x)]^T}{(1+R)^T - 1}$$

= Δgain / gain = (Without Tax − With Tax) / Gain W/o Tax

After-tax **inflation-adjusted** future value of a $1 investment (accrual taxes):

$$FV_{AT,\,infl} = \left[\frac{1+R(1-t_x)}{1+infl}\right]^T = \frac{FV_{AT}}{(1+infl)^T}$$

→ inflat rate carry @ each period
→ let inflat & then compounding

Deferral of Taxes on Capital Gains

After-tax future value of a $1 investment, with deferred capital gains taxed at t_{CG} and the cost basis, B, expressed as percentage of the current market value of the investment:

$$FV_{CG} = (1+R)^T(1-t_{CG}) + \boxed{t_{CG} \times B}$$ OR $= PV_{cg}[(1+r)^T(1-t_{cg}) + t_{cg}]$

Rem: to add back

$$\boxed{Tax\ drag = \frac{(1+R)^T - [(1+R)^T(1-t_{CG}) + t_{CG} \times B]}{(1+R)^T - B} = t_{CG}}$$

i.e. FV with defrd taxes calculated differently

Note equivalence

With Def Tax After-tax **inflation-adjusted** future value of a $1 investment:

added & adjust in Dnr. → to get gain over cost basis.

$$FV_{CG,\,infl} = \frac{(1+R)^T(1-t_{CG}) + t_{CG} \times B}{(1+infl)^T} = \frac{FV_{cg}}{(1+infl)^T}$$

Tax loss harvesting (TLH): pushing portion of tax liab to future years by recognizing incurred loss early helps accumlt more wealth.

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Learning Module 5: Portfolio Management for Institutional Investors

Defined Benefit Pension Plan

$$\text{Funded ratio} = \frac{\text{Fair value of plan assets}}{\text{Present value of defined benefit obligations}}$$

$$\text{Funded status} = \text{Fair value of plan assets} - \text{Present value of defined benefit obligations}$$

Spending Policies (Endowment and Foundation)

Constant Growth rule:

$$Spending_{t+1} = Spending_t \times (1 + Inflation\ Rate)$$

Market Value rule:

$$Spending_{t+1} = Spending\ rate \times Average\ AUM$$

Hybrid rule:

$$Spending_{t+1} = w \times \text{Constant Growth rule} + (1-w) \times \text{Market value rule}$$

Bank and Insurance

Duration of equity capital, D_E

$$D_E = \left(\frac{A}{E}\right) D_A - \left(\frac{A}{E} - 1\right) D_L \left(\frac{\Delta i}{\Delta y}\right)$$

$$\sigma^2_{\Delta E/E} = \left(\frac{A}{E}\right)^2 \sigma^2_{\Delta A/A} + \left(\frac{A}{E} - 1\right)^2 \sigma^2_{\Delta L/L} - 2\left(\frac{A}{E}\right)\left(\frac{A}{E} - 1\right)\rho\sigma_{\Delta A/A}\sigma_{\Delta L/L}$$

$$\text{Equity capital ratio} = \frac{E}{A}$$

D_A = Duration of assets

D_L = Duration of liabilities

i = Effective yield on liabilities

y = Effective yield on assets

$\sigma^2_{\Delta E/E}$ = Variance of change in value of equity capital

$\sigma^2_{\Delta A/A}$ = Variance of change in value of assets

$\sigma^2_{\Delta L/L}$ = Variance of change in value of liabilities

ρ = Correlation between percentage value changes of assets and liabilities

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Sched Algo: suitable when order size is 5-10% of ADV. e.g. {Pctg of Vol (POV), VWAP, TWAP (time wtd)}; exec is relatively liq or order is part of a risk-balanced basket & trading all orders @ a pace will maintain risk balance.
Liq Seeking Algo: for large orders w. quick execution & low impact; chance of adv. reveal due to info leakage.

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(exclude outliers, but VWAP does not.)

Learning Module 6: Trading Costs and Electronic Markets

a) $$\begin{matrix}\text{Effective spread} \\ \text{transaction cost estimate}\end{matrix} = \text{Trade size} \times \begin{cases}\text{Trade price} - \text{Midquote price} & \text{for buy orders} \\ \text{Midquote price} - \text{Trade price} & \text{for sell orders}\end{cases}$$

(mult. by trade size but not 2.)

$$\text{Effective spread} = \begin{cases}2 \times (\text{Trade price} - \text{Midquote price}) & \text{for buy orders} \\ 2 \times (\text{Midquote price} - \text{Trade price}) & \text{for sell orders}\end{cases}$$

where:

$$\text{Midquote price} = \frac{\text{Bid} + \text{Ask}}{2}$$

b) $$\begin{matrix}\text{VWAP transaction} \\ \text{cost estimate}\end{matrix} = \text{Trade size} \times \begin{cases}\text{Trade VWAP} - \text{VWAP benchmark} & \text{for buy orders} \\ \text{VWAP benchmark} - \text{Trade VWAP} & \text{for sell orders}\end{cases}$$

(vol wtd avg px → {other trades + this trade})

Learning Module 7: Case Study in Portfolio Management: Institutional (SWF)

No formula.

Santiago Princ: Prin & Practices for SWF; operational indep, transparency, acct'blty; conduct regular compliance reviews.
Hybrid spending rule by larger endowmt to smooth out spending over mkt cycles; using pure MV → s.d. fluctuat' over yrs
Budget stabi Fund: insulate fiscal budget from commdty Pr ↑↓; risk mgmt; low prob of -ve ret' (1mm); inv in govt bonds.
Dev Fund: eco dev; infra; real ret' target → inc real GDP growth & productivity.
Savings Fund: Long Invm Horizon, intergenerational. Inv in risky, illiq assets - equity+Alts. Earn illiq prem.

Benchmarks: investable; use free-float adj (≠ founder, govt share). Buffering mgmt ⇒ one stock to be in any one segment at a time; Packeting: allows partial holding of a stock in 2 indices simult.

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VOLUME 3 – PERFORMANCE MEASUREMENT

Learning Module 1: Portfolio Performance Evaluation

$$Arithmetic\ excess\ return = R - B$$ = Active Ret"

$$Geometric\ excess\ return = \frac{R-B}{1+B}$$ = Arith Excess ret" / 1+B

Attribution based on Factor Models

$$Active\ Return = Return\ from\ factor\ tilt + Security\ selection\ return$$

$$R - B = \sum (\beta_{P,i} - \beta_{B,i}) \times F_i + Security\ selection\ return$$

where:

$\beta_{P,i}$ = Sensitivity of the portfolio to the given factor

$\beta_{B,i}$ = Sensitivity of the benchmark to the given factor

F_i = Factor return

Equal Wtd = rebal to hold eq wt → better performing stocks are sold.

MktCapWtg: minimal rebal.

relevant pfmnce attrib" used to determine if drivers of mgmt ret" are consistent with the stated investm process.

Micro and Macro Return Attribution

$$\text{Arith. Excess Ret"}\ R - B = \sum_{i=1}^{n} A_i + \sum_{i=1}^{n} S_i + \sum_{i=1}^{n} I_i = \text{Allocat"} + \text{Selection} + \text{Interaction}$$

where:

w_i = Weight of asset i in portfolio

W_i = Weight of asset i in benchmark

R_i = Return of asset i in portfolio

B_i = Return of asset i in benchmark

n = # of asset classes.

Video: https://youtu.be/yrzTVlfqloM

Brinson-Fachler Model

Allocation effect: $A_i = (w_i - W_i)(B_i - B)$ = Δw $(B_i - B)$ → total blm ret" so if $A_i > 0$ & $(B_i - B) >$ ⇒ $(w_i - N_i) > 0$ ⇒ ov wt.

Selection effect: $S_i = W_i(R_i - B_i)$ // keep blm wt.

Interaction effect: $I_i = (w_i - W_i)(R_i - B_i)$ = $\Delta w_i \times \Delta r_i$

Holdings based attrib" analy: – passive strat; little turnover; 17; only snapshot not reflect pfo going fwd

Returns based " ; compare risk chars/drivers agnst style; compare mngr to style indexes – can be applied across Mngrs.

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Brinson-Hood-Beebower Model (BHB)

Allocation effect: $A_i = (w_i - W_i)B_i$ → not taking diff of B_i with overall B ret, ≠ B-F Model

Selection effect: $S_i = W_i(R_i - B_i)$ → same as B-F Model.

Interaction effect: $I_i = (w_i - W_i)(R_i - B_i)$ → " " " $= \Delta w_i \times \Delta r_i$

Decomposing Portfolio Returns

$$P = M + S + A$$
$$= M + (B - M) + (P - B)$$
$$= P$$

where:

P = Portfolio return

M = Market index return

B = Benchmark return

$S = B - M$ = Style return (Misfit active return) = nominal BM ret^n − investor BM ret^n

$A = P - B$ = Active return (True active return) like R-B @ IF

For a good BM, correlat^n b/w active mgmt ret^n (A) vs style ret^n (S) shld be statistically > 0. ⇒ Mngr activ select^n decision aligns with the style earn

Performance Appraisal

Sharpe Ratio w.r.t risk-free rate; Adding infreq trading assets ⇒ Vol ↓ ⇒ SR ↑ → due to smoothing

$$Sharpe\ ratio = \frac{R_p - r_f}{\sigma_p}$$

where:

R_p = Portfolio return

r_f = Risk-free rate

σ_p = Portfolio standard deviation

✓ Treynor Ratio – w.r.t systematic risk from β in Denom.

$$Treynor\ ratio = \frac{R_p - r_f}{\beta_p}$$

where:

R_p = Portfolio return

r_f = Risk-free rate

β_p = Portfolio beta (Systematic risk)

$$\beta = \rho_{stock,mkt} \times \frac{\sigma_{stock}}{\sigma_{mkt}}$$

In a mkt neutral trade with 2 stocks, Buy A for \$ = Tot Pos / P_{xA}. Now to determine # B to short, $= \frac{Tot\ Pos_A}{P_{xB}} \times \frac{\beta_A}{\beta_B}$

$Tot\ Pos_A \times \beta_A = Tot\ Pos_B \times \beta_B$ ← (used to adjust # count)

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Information Ratio with blank.

$$Information\ ratio = \frac{R_p - R_B}{\sigma_{R_p - R_B}}$$

where:

R_p = Portfolio return – over multi periods via GM.

R_B = Benchmark return

$\sigma_{R_p - R_B}$ = Active risk (Tracking risk/Tracking error)

Appraisal Ratio

$$Appraisal\ ratio\ (Treynor\ Black\ ratio) = \frac{\alpha}{\sigma_\varepsilon}$$

where:

σ_ε = Standard error of regression (from factor model)

$$= \sqrt{\sigma_P^2 - \beta_i^2 \sigma_M^2}$$

Jensen $\alpha = R_i - [R_f + \beta_i(R_M - R_f)]$

Sortino Ratio

$$Sortino\ ratio = \frac{R_p - r_T}{\sigma_{Downside}}$$

→ over multiple periods may follow GM like TWR(2); see safety first ratio @ 9; only adj Numer

$$= \frac{E(R) - R_L}{\sigma}$$

where:

r_T = Investor's minimum acceptable return/target return // Threshold ret'n

$$\sigma_{Downside} = \sqrt{\frac{1}{N}\sum_{t=1}^{N} \min(r_t - r_T, 0)^2}$$

→ whenever ret'n falls short of threshold; Use only those ret'n for DD which fall below r_T.

for all periods; not just numerator counts.

Capture Ratio → merge vis blank. if cap Rat > 1 ⇒ CONVEX ret'n profile

$$Capture\ Ratio = \frac{Upside\ Capture}{Downside\ Capture} = \frac{\left[\frac{R_m}{R_B}\right]_{R_B \geq 0}}{\left[\frac{R_m}{R_B}\right]_{R_B < 0}}$$

✓ $Upside\ Capture = \frac{R_m}{R_B}$ when $R_B \geq 0$

✓ $Downside\ Capture = \frac{R_m}{R_B}$ when $R_B < 0$ If DC<1, outperformers will continue to outperform. If DC>1, inline performers ⇒ now underperform.

where:

R_m = Manager's return

R_B = Benchmark return

See also: Safety First Ratio @ Pg 9.

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Maximum Drawdown

$$Maximum\ Drawdown = \min\left(\left[\frac{V(m,t) - V(m,t^*)}{V(m,t^*)}\right], 0\right)$$

→ divided by peak val

where:

$V(m,t)$ = portfolio value of manager m at time t

$V(m,t^*)$ = peak portfolio value of manager m [at $t-t^*$]

$t > t^*$ for periods after peak value

Video: https://youtu.be/0tRDDT9E9AU

Learning Module 2: Investment Manager Selection

Performance-Based Fees → does not include Std fees → which is outside all this.

If manager is fully exposed to upside and downside:

Computed fee = Base fee + Sharing of performance

If manager not exposed to downside:

Computed fee = Higher of either [1] Base fee, OR
[2] Base fee plus sharing of positive performance

If manager is not exposed to downside, and there is a maximum fee.

Computed fee = Higher of either [1] Base fee, OR
[2] Base fee plus sharing of performance, to a maximum fee

Hi Wtr Mark – 1) prevent mngr from incentive fees until fund ret" to prev. peak 2) If way below, mngr incentivized to ret" funds to investors i/o continue w'pfm.

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GIPS allows creating min threshold to include pfo into composite; likewise pfo breach floor/ter before exclusion

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Learning Module 3: Overview of the Global Investment Performance Standards

Time-weighted return (TWR) – not affected by timing of cash inflows ⇒ ∴ relevant when exit mngr can't control timing

$$r_{twr} = \left[(1 + r_{t,1}) \times (1 + r_{t,2}) \times \ldots \times (1 + r_{t,n}) - 1\right]$$ → get ann. TWR: $\left\{(1 + r_{twr})^{1/n} - 1\right\}$

$r_{t,1}$ *through* $r_{t,n}$ = Sub-period returns – after deducting c/f at end of yr; +ve vals into are outflows

after 1 Jan 2005 ← **Modified Dietz method** = Aggregate Return Method for Composite.
Before " " → Original Dietz method

$$r_{ModDietz} = \frac{V_1 - V_0 - CF}{V_0 + \sum_{i=1}^{n}(CF_i \times w_i)} = r_{comp}$$

Video: https://youtu.be/guZWVXirJL0

Modified IRR method ≃ MWR (Money Wtd Ret) = find ann. ret° that equates terminal val of fund.

Final val of fund, $V_1 = \sum_{i=1}^{n}[CF_i \times (1 + MIRR)^{w_i}] + \underbrace{V_0(1 + MIRR)}_{\text{extra}}$

$V_1 = \sum CF_i(1 + MWR)^n$ where $CF_1 = V_1$ i.e. initial val; subseq, CF_i = cashflow.

where:

w_i = Proportion of period (in days) that each cash flow has been in the portfolio

$$= \frac{CD - D_i}{CD}$$

CD = Total number of calendar days in the period

D_i = Number of calendar days from the beginning of the period to the time cash flow CF_i occurs

$$CF = \sum_{i=1}^{n} CF_i$$

Money Wtd Ret° (MWR): when mngr can control timing of ext c/f; pfo are closed end, fixed life or with sig. illiq inv. ≃ IRR, s/t PV(in) = PV(outflo)

Composite Time-Weighted Return i.e. $0 = \sum \frac{c/f}{(1+IRR)^t}$

Asset-weight individual portfolio returns using **beginning-of-period** values

$$r_C = \sum_{pi=1}^{n} r_{pi} \times \frac{V_{0,pi}}{\sum_{pi=1}^{n} V_{0,pi}}$$

where:

r_C = Composite return

r_{pi} = Return of an individual portfolio i

$V_{0,pi}$ = Beginning value of portfolio i

$\sum_{pi=1}^{n} V_{0,pi}$ = Total beginning fair value of all individual portfolios in the composite

GIPS Compo. Report: a) 10-yr ann ret° for compo. & its b/m; b) 3-yr annualized ex-post s/d of monthly ret° for compo. & b/m; c) Tot val of ⊕ in comp. & AUM at yr end; d) # of pfo & reqd if # pfo ≥ 6 e) Int. dispersion is reqd if # pfo ≥ 6.

Firm can be defined for GIPS based on fund autonomy, discretion over AUM, complete investment decision authority.

GIPS – must include ret° from cash & cash equiv in pfo for all tot ret° calcn. Changing meth of calcn (TWR vs MWR) needs to be disclosed.

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Use a method that reflects both beginning-of-period values and external cash flows

$$r_C = \sum_{pi=1}^{n} r_{pi} \times \frac{V_{pi}}{\sum_{pi=1}^{n} V_{pi}}$$

where:

V_{pi} = Beginning value of portfolio i + Weighted external cash flows

$\sum_{pi=1}^{n} V_{pi}$ = Total beginning fair value and weighted external cash flows of all individual portfolios in the composite

Aggregate Return Method

$$r_{ModDietz} = \frac{V_1 - V_0 - CF}{V_0 + \sum_{i=1}^{n}(CF_i \times w_i)}$$

Numer is linear change

Denom is total sum (original base)

where:

V_1 = Ending value of composite

V_0 = Beginning value of composite

$$CF = \frac{Total\ of\ external}{cash\ flows\ in\ composite} = \sum_{i=1}^{n} CF_i$$

$$\sum_{i=1}^{n}(CF_i \times w_i) = \frac{Total\ of\ \mathbf{weighted}\ external}{cash\ flows\ in\ composite} = \sum_{pi=1}^{n} V_{pi}$$

Equal-weighted standard deviation

$$S_c = \sqrt{\frac{\sum_{i=1}^{n}(r_i - \bar{r}_c)^2}{n}}$$

where:

r_i = return for portfolio i

$\bar{r}_c$ = equal-weighted mean

n = number of portfolios in composite

Asset-weighted standard deviation

$$S_{c,aw} = \sqrt{\sum_{i=1}^{n} w_i \times \left(r_i - \bar{r}_{proxy}\right)^2}$$

$\bar{r}_{proxy}$ = asset-weighted mean return $= \sum_{i=1}^{n} w_i \times r_i$

w_i = weight of portfolio $i = \frac{V_{0,i}}{V_{0,Total}}$

In an o/wise GIPS compliant presentation that complies with local laws that conflict with GIPS reqmt, firms must disclose this fact & disclose the manner in which the laws conflict with GIPS reqmt.

Valuation hierarchy from GIPS: -1) Quoted Px for ~ Instrm in active mkt, 2) Quotes for Instrm in inactive mkt, 3) Mkt-based input 4) Subjective, unobservable inputs.

Buying Call Opt → ... ; Buying Put Opt ⇒ Eff Dur ↓ & Convexity

CFA Level 3 (2025) Formula Sheet by Fabian Moa
Portfolio Management Pathway

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VOLUME 4 – DERIVATIVES AND RISK MANAGEMENT

Learning Module 1: Option Strategies

positive ← Δ of call ∈ [0,1]; 10-delta call ⇒ 0-10 ITM
10-delta is far OTM vs 20-Delta (more expensive)
↳ lower prob. of going ITM.
Whereas Put Delta = [-1, 0] ⇒ Negative

Put-call parity

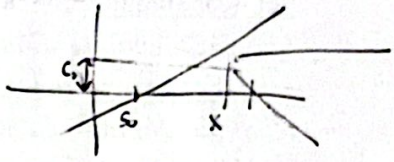
$$S_0 + p_0 = c_0 + \frac{X}{(1+r)^T}$$

Put-call-forward parity
replace spot with F_0 - equiv

$$\frac{F_0}{(1+r)^T} + p_0 = c_0 + \frac{X}{(1+r)^T}$$

Synthetic long forward ← (rearrange)

$$c_0 - p_0 = \frac{F_0 - X}{(1+r)^T}$$



Option premium = Time value + Intrinsic value ⇒ S − C

Covered Calls long stock & short call; Video: https://youtu.be/2SocH6PqhOk

$Expiration\ value = S_T - Max(S_T - X, 0)$
$Profit\ at\ Expiration = S_T - Max(S_T - X, 0) + c_0 - S_0$ → push p's
$Maximum\ gain = X - S_0 + c_0$ → i.e. $S_T - S_0$ → prem received due to writing call → $= S_T - X$
$Maximum\ loss = S_0 - c_0$
$Breakeven\ price = S_0 - c_0$

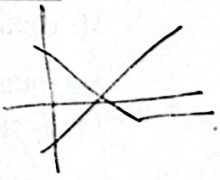
sell @ T

$Position\ delta = Stock\ Delta - Call\ Delta = 1 - \Delta_{call}$ ∵ stock − call
$Position\ gamma = Stock\ Gamma - Call\ Gamma = -\text{Gamma}_{call} < 0$
$Position\ vega = Stock\ Vega - Call\ Vega = -\text{Vega}_{call} < 0$
$Position\ theta = Stock\ Theta - Call\ Theta = -\text{Theta}_{call} < 0$

∴ stock regs, Γ, Θ ≡ 0
Γ, V, Θ are negative

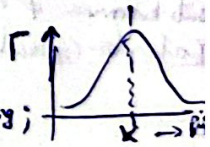
Protective Puts long stock + long put; S+P Video: https://youtu.be/VLK1lXbXtRk

$Expiration\ value = S_T + Max(X - S_T, 0)$
$Profit\ at\ Expiration = S_T + Max(X - S_T, 0) - S_0 - p_0$ → put prem paid
$Maximum\ gain = \infty$
$Maximum\ loss = S_0 - X + p_0$
$Breakeven\ price = S_0 + p_0$



$Position\ delta = Stock\ Delta + Put\ Delta = 1 + \Delta_{put}$
$Position\ gamma = Stock\ Gamma + Put\ Gamma = \text{Gamma}_{put} > 0$
$Position\ vega = Stock\ Vega + Put\ Vega = \text{Vega}_{put} > 0$
$Position\ theta = Stock\ Theta + Put\ Theta = \text{Theta}_{put} < 0$

Γ, V, Θ > 0



Gamma $\Gamma = \frac{\partial^2 V}{\partial^2 S}$; Hi Gamma ⇒ Δ is v. sensitive Δ px. Δ chgs most rapidly near ATM ⇒ req freq hedging;

Cash Covered Put: write put on stock & deposit K into a desig 23 noted a/c. Recv prem. & earn int on covering funds.

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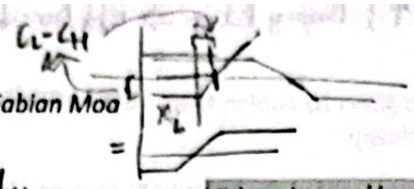

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Call Bull Spread $= Call_L - Call_H$ Video: https://youtu.be/3NHwelzEU0k

$Expiration\ value = Max(S_T - X_L, 0) - Max(S_T - X_H, 0)$
$Profit\ at\ Expiration = Max(S_T - X_L, 0) - Max(S_T - X_H, 0) - (c_L - c_H)$ → prem recvd; → prem paid
$Maximum\ gain = X_H - X_L - (c_L - c_H)$
$Maximum\ loss = c_L - c_H$ → net prem paid
$Breakeven\ price = X_L + c_L - c_H$ → low strike + net prem paid

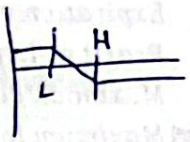
Put Bull Spread $= Put_L - Put_H$ Video: https://youtu.be/Lf1Fi-zy7w4

$Expiration\ value = Max(X_L - S_T, 0) - Max(X_H - S_T, 0)$
$Profit\ at\ Expiration = Max(X_L - S_T, 0) - Max(X_H - S_T, 0) - (p_L - p_H)$
$Maximum\ gain = p_H - p_L$
$Maximum\ loss = X_H - X_L + p_L - p_H = X_H - X_L - (p_H - p_L)$
$Breakeven\ price = X_H + p_L - p_H$

Bear spread: buy higher strike & sell lower strike.
Bull " : " lower " " higher "

Put Bear Spread $= Put_H - Put_L$ Video: https://youtu.be/eTejezNXZbU

$Expiration\ value = Max(X_H - S_T, 0) - Max(X_L - S_T, 0)$
$Profit\ at\ Expiration = Max(X_H - S_T, 0) - Max(X_L - S_T, 0) - (p_H - p_L)$
$Maximum\ gain = X_H - X_L - (p_H - p_L)$
$Maximum\ loss = p_H - p_L$
$Breakeven\ price = X_H - p_H + p_L$



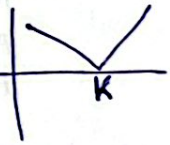
Call Bear Spread $= Call_H - Call_L$

$Expiration\ value = Max(S_T - X_H, 0) - Max(S_T - X_L, 0)$
$Profit\ at\ Expiration = Max(S_T - X_H, 0) - Max(S_T - X_L, 0) - (c_H - c_L)$
$Maximum\ gain = c_L - c_H$
$Maximum\ loss = X_H - X_L + c_H - c_L$
$Breakeven\ price = X_L + c_L - c_H$

Straddle
- short = $-Call_K - Put_K$ → recv 2 prem → expect low vol.
- long = $+Call_K + Put_K$ → pay 2 prem → exp. hi vol

Video: https://youtu.be/oDklmeMTnCg

Long straddle



$Expiration\ value = Max(S_T - X, 0) + Max(X - S_T, 0)$
$Profit\ at\ Expiration = Max(S_T - X, 0) + Max(X - S_T, 0) - c_0 - p_0$
$$Maximum\ gain = \begin{cases} \infty & S_T > X \\ X - S_T - c_0 - p_0 & S_T < X \end{cases}$$
$Maximum\ loss = c_0 + p_0$
$Breakeven\ price = X \pm (c_0 + p_0)$

Long cal. spread: Buy long-dated op & sell short-dated op; capture impact of time decay; Neutral outlook + stable IV
straddle also work in neutral outlook; but long straddle when IV exp ↑↑ & short straddle if expect imp vol (IV). to ↓↓
All or nothing option = fixed payout if it touches K @any time b4 expiry; Knock-in: plain vanilla option when spot touches barrier
But barrier ≠ ex. price (K). Knock out: ceases to exist when spot touches barrier.
Calendar Spread: short on near-dated call opn & going long on same longer-dated call opn. // Long calendar spread position

Collar when owning stock while maintaining some upside. Video: https://youtu.be/LkS_sxmg2cs for profit, offset the price of put for protection by selling a call.

Note: $X_L < S_0 < X_H$

$$Expiration\ value = S_T + Max(X_L - S_T, 0) - Max(S_T - X_H, 0)$$
$$Profit\ at\ Expiration = S_T + Max(X_L - S_T, 0) - Max(S_T - X_H, 0) - S_0 - p_0 + c_0$$
$$Maximum\ gain = X_H - S_0 - p_0 + c_0$$
$$Maximum\ loss = -X_L + S_0 + p_0 - c_0$$
$$Breakeven\ price = S_0 + p_0 - c_0$$

For zero-cost collar, $p_0 = c_0$

Implied Volatility

For 252 trading days in a year and 21 trading days in a month:

$$\sigma_{annual} = \sigma_{monthly}\sqrt{\frac{252}{21}}$$

i.e. $\frac{\sigma_a}{\sqrt{T_a}} = \frac{\sigma_m}{\sqrt{T_m}}$

Delta hedging

$$N_{position} \times \text{Delta}_{position} + N_{hedge} \times Delta_{hedge} = 0$$

Learning Module 2: Swaps, Forwards, and Futures Strategies

Managing Interest Rate Risk

Interest Rate Swaps

$$N_S = \left(\frac{MDUR_T - MDUR_P}{MDUR_S}\right) MV_P$$

N_S = Swap notional principal
$MDUR_T$ = Target modified duration
$MDUR_P$ = Modified duration of portfolio
$MDUR_S$ = Modified duration of swap
MV_P = Market value of portfolio

Note: Modified duration of cash = 0 (unless stated otherwise in case)

Money Market Instrument

$$BPV = Face\ Value \times \frac{Days}{360} \times 0.01\%$$

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Duration Neutral curve Exposure: Hedge Ratio = $\frac{DV01\ (long)}{DV01\ (short)}$

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Interest Rate Futures

ED Fu ✓ $Eurodollar\ futures\ price = 100 - Annualized\ yield$

Treasury Futures

$$Principal\ invoice\ amount = \frac{Futures\ settlement\ price}{100} \times Conversion\ Factor \times Contract\ size$$

Basis Point Value Hedge Ratio (BPVHR)—To synthetically rebal a pfo to a target ModDur using bond index futures.

$$BPVHR = \left(\frac{BPV_T - BPV_P}{BPV_{CTD}}\right) \times Conversion\ Factor$$

Mod Durn. $BPV_T = MDUR_T \times 0.01\% \times MV_P$
$BPV_P = MDUR_P \times 0.01\% \times MV_P$
$BPV_{CTD} = MDUR_{CTD} \times 0.01\% \times MV_{CTD}$

Rem: convert to bps; mDur used, viz diff from MMF @ ≥5 → If rates drop/chng → recal. MV with new spread → New BPV

$$MV_{CTD} = \frac{CTD\ price}{100} \times Futures\ contract\ size$$

$BPV_F = \frac{BPV_{CTD}}{CF}$ — already in the formula

Hedging Currency Risk with Futures

$$N_f = \frac{Required\ currency\ hedge}{Contract\ size}$$

Hedging Equity Risk with Futures

eg. # of contr needed to equitize the cash plan.; May need to use in 2 steps for cash to go from 0 to x & then existing pfo @ β=x+ New cash to then go to β=y.

$$N_f = \left(\frac{\beta_T - \beta_S}{\beta_f}\right)\left(\frac{S}{F}\right)$$

where:

β_T = Target beta = 1 for index
β_S = Portfolio beta = 0, if cash pfo.
✓ β_f = Futures beta
S = Portfolio market value
F = Futures price

$$Effective\ beta = \frac{\%\Delta Portfolio\ value}{\%\Delta Index\ value}$$, like duration

✓ *Note*: Beta of cash = 0 → so transaction from equity to bond will go via cash with $\beta_f = 0$.

Video: https://youtu.be/VMVQ2GOrG0Q

Trading Fwd Rate Bias = buy ccy selling @ Fwd discount & selling ccy trading @ Fwd premium. If trading against Fwd Rate Bias (eg. selling a fwd discount ccy) → may under hedge.

Diff of Var Swap with VIX fut = 1) convex, 2) vol. used to cal. payoff, 3) Struc of VIX curve. As realized vol ↑ ⇒ more payoff T. → used to take dir. bet on implied vs realized vol.; VIX fut: views on timing & mag of Δimplied.

Variance Swap ≡ long a basket of options & short the u/l. It's long gamma with convex payoff profile.

At maturity, → $\text{Settlement amount}_T = \text{Vega notional}\left(\frac{\sigma^2 - X^2}{2X}\right)$ ← strike price (in %, not decimal)

$= \text{Variance notional}(\sigma^2 - X^2)$

Note diffn btw vega notional vs var notional.

∴ Var Not = $\frac{\text{Vega Not}}{2 \times \text{strike}} = \frac{100\,k}{2 \times 25}$

If settlmt amt > 0, buyer gets money.

$$\text{Value of long variance swap at time } t = \frac{\text{Variance notional} \times (\sigma_t^2 - X^2)}{1 + r \times \left(\frac{T-t}{T}\right)}$$ → linear discounting rate interpolated

where:

$$\sigma_t^2 = \frac{t}{T} \times \underbrace{\sigma^2_{realized}(0,t)}_{\text{historical}} + \frac{T-t}{T} \times \underbrace{\sigma^2_{implied}(t,T)}_{\text{future implied}}$$ <wtd. Avg>; dur at T (settlmt), σ^2 = realized var

Video: https://youtu.be/YVNPCXGTdWk

Probability of a Change in Federal Funds Rate

$$\text{Probability of a change in federal funds rate} = \frac{\text{Implied Fed funds rate} - \text{Current fed funds rate}}{\text{Target rate hike}}$$

$\text{Current fed funds rate} = \underline{\text{Midpoint}}\text{ of current target range}$

Fed funds futures price = 100 – Implied Fed funds rate

Learning Module 3: Currency Management An Introduction

$$R_{DC} = (1 + R_{FC})(1 + R_{FX}) - 1 \longrightarrow (A)$$

R_{DC} = Domestic-currency return
R_{FC} = Foreign-currency return
R_{FX} = Percentage change in foreign currency against domestic currency (currency quoted as DC/FC) → see @12

Note

Video (Unhedged Returns): https://youtu.be/7Cycb5teSbU

From (A), **Volatility of foreign asset (in domestic currency terms)**

$$\sigma^2_{DC} = \sigma^2_{FC} + \sigma^2_{FX} + 2\sigma_{FC}\sigma_{FX}\rho_{FC,FX}$$ ← linked to the ret' equat' above

Rem to use w_1, w_2 as ... esp. for a long-short p/f

For multi ccy p/fo

For a foreign-risk free asset: → only exchange rate risk remains → but use R_{FC} to convert σ_{FX} to σ_{DC}

$$\sigma_{DC} = \sigma_{FX}(1 + R_{FC})$$ → convert foreign ccy ret' to dom. ccy

Fwd Rate Bias: fwd @ prem to spot; biased predictor of future spot rate. Overestimates fut spot rate when fwd is @ p...

↳ + R_{FC} becomes const in (A) & leaving SD (R_{FX}

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Minimum Variance Hedge Ratio (MVHR)

$$y_t = \alpha + \beta x_t + \varepsilon_t$$

where:

y_t = change in value of asset to be hedged (measured in DC) = R_{DC} ✓

x_t = change in value of hedging instrument (measured in DC) = R_{FX} ✓

$$MVHR, \quad \beta = \frac{Cov(x,y)}{Var(x)} = \rho_{x,y}\left(\frac{\sigma_y}{\sigma_x}\right)$$; ≈ as 'β' like stock $\beta = \left(\frac{\rho\sigma_i}{\sigma_m}\right)$

Contrarian : bottom-up; based on indiv stocks

Non-Deliverable Fwd (NDF): cred risk is lower than outright fwd, ∵ princ sums do NOT move, unlike vanilla fwd. NDF only requires payment/receipt of net position. Regular fwd requires phy. exchng. of ally princ. NDF are cash settled in non-controlled ccy of ccy pair i/o phy. settlmt. The ctrlled ccy is neither delivered nor recvd.

Contrary to UIRP, (carry trades) are profitable. Hy yld ccy tends to see ccy ↑ & lower-yld funding ccy ↓.

Full replica^{n}: ↓ Enhan. Dim by holding cash if ER on index < risk-free rate. b) Uptrend MM: i) mgmt fee > 0; ii) tran costs (iii) cash drag.

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PORTFOLIO MANAGEMENT PATHWAY

Learning Module 1: Index-Based Equity Strategies Index shld be rules-based, transparent & investable.

Attribution Analysis

$$\text{Contribution of sector } i \text{ to portfolio return} = \text{Weight of sector } i \text{ in portfolio} \times \text{Return of sector } i$$

$$\text{Contribution of sector } i \text{ to benchmark return} = \text{Weight of sector } i \text{ in benchmark} \times \text{Return of sector } i$$

Learning Module 2: Active Equity Investing: Strategies

Growth at a Reasonable Price (GARP)

$$PEG\ ratio = \frac{P/E}{g\ (in\ \%)}$$

If PEG < 1 ⇒ undervalued ∴ BUY
If " > 1 ⇒ overvalued

Information Coefficient

Pearson IC = Correlation between factor score and subsequent month return $\text{Corr}(r_{t+1}, F_t)$

Spearman rank IC = Correlation between rank of factor score and rank of subsequent month return $\text{Corr}[R(F_t); R(r_{t+1})]$

Returns-based Style Analysis (RBSA)

$$r_t = \alpha + \sum_{s=1}^{m} \beta^s R_t^s + \varepsilon_t$$

Portion of ret^{n} attrib to $(\alpha+\epsilon)$ is much greater in a conc. pfo.

where:

r_t = fund return within period ending at time t
R_t^s = return of style index s in same period
β^s = fund exposure to style s ($\sum_{s=1}^{m} \beta^s = 1$; $\beta^s > 0$ for long-only)
α = manager's value added
ε_t = residual return that cannot be explained by the styles used → Idiosync. Risk

(p. 34)

Equity Index Swap → specialized blnk exposure: pay float & recv indx ret^{n}; Equity Swap = pay float & recv tot ret^{n} of equity

Factor-mimicking Pfo: is $ neutral L/S pfo that provides pure exposure to factor & no other. High cost due to 1) Rs in large # stocks w/o considering tran cost; 2) No check for borrowing stocks @ reasonable cost?

Factor-tilting pfo: only long; Hedged pfo: include shorts.; More conc posn. while Factor-tilting - factor exp/diversif/Mini

Large Cap Mandate = M-cap > $10bn; Small Cap = M-cap ∈ [$300mn to $10bn]

FF Hedged Factor pfo: i) don't elim the eff of other factors ∵ not pure factor pfo; info abt mid quartiles wasted; stocks maybe upc

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Wider dispersion in exp. ret^n btw 2 mgrs → costs of Type I or II error ↓.

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Learning Module 3: Active Equity Investing: Portfolio Construction

= security selection + factor timing

Active Return = Rewarded Factors + α (Skill) + Idio ret^n

$$R_A = \sum_{i=1}^{N} \Delta W_i R_i$$

where: R_i = return on security i

ΔW_i = active weight = $W_{pi} - W_{Bi}$

$$R_A = \sum_{i=1}^{N} (\beta_{pk} - \beta_{bk}) \times F_k + (\alpha + \varepsilon)$$

pfo - bm

where:

β_{pk} = sensitivity of portfolio (p) to rewarded factor (k)

β_{bk} = sensitivity of benchmark (b) to rewarded factor (k)

F_k = return of each rewarded factor

α = active return that can be attributed to manger's specific skill/strategies (① security selection, ② factor timing)

ε = idiosyncratic return

New

$$Portfolio\ turnover\ ratio = \frac{Min(Purchases, Sales)}{Average\ monthly\ net\ assets}$$

Fundamental Law of Active Management on expected Active Return

$$E(R_A) = IC \times \sqrt{BR} \times \sigma_{R_A} \times TC$$

where:

IC = Information coefficient If $E(R_A)$ < annualized actual R_A ⇒ stated IC must be lower than realized IC.

BR = Breadth (Number of independent decisions made per year) Rem

TC = Transfer coefficient (Unconstrained portfolio → TC = 1) ⇒ ability to translate pfo insight → Investment decision

σ_{R_A} = Manager's active risk

$$Active\ Share = \frac{1}{2}\sum_{i=1}^{n} |w_{p,i} - w_{B,i}|$$

— will be hi if idio risk is hi; If pfo uses factor select^n ⇒ target factor expo ⇒ Min. idio.

where: n = number of securities in either portfolio or benchmark

C

$$Active\ Risk\ or\ Tracking\ Error, \sigma_{R_A} = \sqrt{\frac{\sum_{t=1}^{T}(R_{At})^2}{T-1}}$$

①

Val at Risk (VaR): Min loss that can be expected x% of time over a specified p'd. CVaR = loss expected if VaR hold is exceeded.

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Active risk or TE $\sigma_{R_A}^2 = \sigma^2\left(\sum_{i=1}^{N}(\beta_{pk} - \beta_{bk}) \times F_k\right) + \sigma_\varepsilon^2$ → (2)

factor expo. + idio

$\sigma^2\left(\sum_{i=1}^{N}(\beta_{pk} - \beta_{bk}) \times F_k\right)$ = Variance attributed to factor exposure

σ_ε^2 = Variance attributed to idiosyncratic risk

Absolute Risk Attribution Video: https://youtu.be/zpk24jsMGDM

$$V_p = \sum_{i=1}^{n}\sum_{j=1}^{n} x_i x_j C_{ij}$$ → 2 summation over i & j

Asset i contrib to Tot Var → $$CV_i = \sum_{j=1}^{n} x_i x_j C_{ij} = x_i C_{ip}$$ only 1 summation over j ∴ CV_i

where:

V_p = Portfolio variance

CV_i = Contribution of asset i to portfolio variance

x_i = weight of asset i in portfolio

C_{ij} = Covariance of returns between asset i and j

C_{ip} = Covariance of returns between asset i and portfolio $= \sum_{j=1}^{n} x_j C_{ij}$

Relative Risk Attribution – wrt benchmark.

$$AV_p = \sum_{i=1}^{n}\sum_{j=1}^{n}(x_i - b_i)(x_j - b_j)RC_{ij}$$

$$CAV_i = \sum_{j=1}^{n}(x_i - b_i)(x_j - b_j)C_{ij} = (x_i - b_i)C_{ip}$$

where:

AV_p = Variance of portfolio's active return. wrt bmark

CAV_i = Contribution of asset i to portfolio active variance

b_i = benchmark weight in asset i or & likewise for j

RC_{ij} = Covariance of relative returns between asset i and j ⇒ ~~RC~~ $RC_{ij} \overset{Cov}{=} [(R_i - B)(R_j - B)]$

C_{ip} = Covariance of relative returns between asset i and portfolio

(C) cov. of relative ret" ? ✓

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Equity Mkt Neutral: don't capture mkt beta → ∴ struggle in strong bull mkt with low dispersion. Net zero mkt exposure & minimize systematic risk. Gen. ∝ to stock selection via idio risk, not systematic risk.

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P

Compounded Return (Geometric Return)

$$R_g = k \times R_a - (k-1)R_d - \frac{1}{2}(k \times \sigma)^2$$

R_g = Compounded/geometric return
R_a = Arithmetic/periodic return
k = Leverage factor = $Asset/Equity$
R_d = Cost of funding leverage

Note **Gross and Net Exposure** Unconstrained L/S pfo has greater risk & ret° opportunities than a constrained L/S pfo, eg 130/30

Equity L/S fund cannot play role of cap. growth in multi-Asset class pfo. But can provide non. risk diversifcan.

✓ $Gross\ Exposure = Long\ positions + |Short\ positions|$; Gross Leverage = Gross Exposure ÷ NAV.

✓ $Net\ Exposure = Long\ positions - |Short\ positions|$

Long Extension = Net +ve Beta; Short bias = Net -ve Beta.

Learning Module 4: Liability-Driven and Index-Based Strategies

Cash ≃ zero duration, inflation-protected Ⓐ?

Macaulay Duration, Dispersion, and Convexity Immunized pfo must have Ⓐ that satisfys :-

match to immu ← $Macaulay\ Duration = \sum_{t=1}^{N} w_i \times t$ → Ⓐ a) MacDur of Ⓐ = liab due dt (MacDur (L)) b) $PV_A \geq PV_L$ c) Min pfo Convx.

against small ⊥ ylc shift d) if Δclf yld on Ⓐ ≡ Δclf yld on Ⓑ ⇒ immunizd to non-⊥ shift to ylc.

<Using a ZCB to defeasance strat to immu pfo. ⇒ 1) No reinvm risk 2) No price risk.>

$$w_i = \frac{PV_i}{PV}$$

Note: To annualize MacDur, divide by periodicity. ⇒ $MacDur_{annual} = \frac{MacDur\ per\ period}{t}$ (→ from clf timing in periods using Ⓐ)

Note

$Dispersion = \sum_{t=1}^{N} w_i \times (t - MacDur)^2$ ∴ dispersion around (macDur)

against large ⊥ ylds ← $Convexity = \frac{1}{(1 + Cash\ flow\ yield)^2} \sum_{t=1}^{N} w_i \times t(t+1)$ → IRR on various clf from compo. bonds. Rate that causes PV of exp. fut clf ≡ current MV.

Note: To annualize dispersion and convexity, divide by periodicity squared Note → Need squaring

For an immunized pfo, ✓ $Convexity = \frac{MacDur^2 + MacDur + Dispersion}{(1 + Cash\ flow\ yield)^2}$ → wtd avg time to receipt of cashflows around MacDur → see also @(35)

Video: https://youtu.be/qeky-p7Hljw

Bullet pfo.: clf conc. around invm Horizon ⇒ lower reinvm risk; Narrower disp of clf → lower convx than ladder or barbell pfo of equivalent durn & clf yld.

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Structural Risk: potential for non-// shifts & twists in y/c → Δcf yld → mismatch btw (A) & (L). Minimized by reducing the or minimizing the dispersion of the bond positions

CFA Level 3 (2025) Formula Sheet by Fabian Moa
Portfolio Management Pathway

FABIAN MOA

Note

$$\text{Modified Duration} = \frac{\text{Macaulay Duration}}{1 + \text{Cash flow yield}/\text{freq}}$$

– captures Δbd due to Δyld, regardless of where Δy curve from, eg. Δspread or Δint rates. Can't identify pfo with shortfall any risks.
→ adjust cf yld with freq of payout;

based on actual money/MV, not par value.

$$\text{Money Duration} = \text{Market value} \times \text{Modified Duration}$$

$$BPV = \text{Market value} \times \text{Modified Duration} \times 0.0001 = \text{Money Dur} \times \frac{1}{10^4}$$

Note: For zero-coupon bonds, Macaulay duration = Maturity

Number of bond futures contracts required to close the duration gap, based on BPV directly:-

$$N_f = \frac{\text{Liability BPV} - \text{Asset BPV}}{\text{Futures BPV}}$$

where:

$$\text{Futures BPV} \approx \frac{BPV_{CTD}}{CF_{CTD}}$$

$$\text{Duration gap} = \text{Asset BPV} - \text{Liability BPV}$$

Video (Derivatives overlay with futures): https://youtu.be/3ZlCA1nP8Zc

Video (Contingent immunization): https://youtu.be/bL9P0j5LNJk

→ requires that $MV_A \geq MV_L$; use (A) cf yld to get PV of (L) ∴ pension plan w. shortfall can't use this. The excess of (A) over (L) can be invested

Required Swap Notional to close the duration gap actively.

$$NP = \frac{\text{Liability BPV} - \text{Asset BPV}}{\text{Swap BPV}/100}$$

Note: Swap BPV quoted per $100 notional → extra, vis-a-vis bond futures contr

Video: https://youtu.be/LGsJEXCYH0g

$$\frac{\text{Asset}}{\text{BPV}} \times \frac{\Delta\text{Asset}}{\text{yields}} + \frac{\text{Hedge}}{\text{BPV}} \times \frac{\Delta\text{Hedge}}{\text{yields}} = \frac{\text{Liability}}{\text{BPV}} \times \frac{\Delta\text{Liability}}{\text{yields}}$$

→ mismatch leads to spread Risk

Bond pfo loses value when rates ↑ Tsy Fut prx ↓ ⇒ short fut posn gains from drop ∴ short fut can hedge IRR.

Duration Times Spread (DTS): pfo DTS = w_1 (~~DTS~~ EffSpDur$_1$ × OAS$_1$) + w_2 (EffSpDur$_2$ × OAS$_2$); & so on.
When (index) spread widens/drops: [ΔOAS/OAS ×100] ⇒ ΔMV of pfo = MV × (− pfo DTS × % chng in OAS); w_1 = $MV_1/(MV_1+MV_2)$ etc. → like ex 35

Duration Matching liability Mandate: 1) Min. risk of shortfall in cash ∵ D^A = PV of liab & cf from coupn payout to offset liab cf.

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Receiver swaption: option to enter recv swap, i.e. recv fxd, pay float; Payer " : " " " payor " i.e. pay fxd, recv float | 'payor' & 'recv' refer to fixed leg in swap.

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Learning Module 5: Yield Curve Strategies

$$\text{Yield curve slope} = \text{Long term yield} - \text{Short term yield}$$

via options: long wings & short body.

Bfly trade = positioning in the bullet & barbell.

$$\text{Butterfly spread} = -\text{Short term yield} + 2 \times \text{Medium term yield} - \text{Long term yield}$$

$= -s + 2m - l$; Inc in Bfly spread ⇒ St & Lt ylds dec relative to med term. ∴ long bonds @ wings & short bullet

$$\%\Delta PV^{Full} = -(ModDur \times \Delta Yield) + \frac{1}{2} \times Convexity \times (\Delta Yield)^2$$

$$= \%\Delta PV^{Full} \times PV^{Full}$$

Butterfly shift: if +ve ⇒ non // shift in y/c: short & long rates inc by more than mid rates. ∴ dec curvature → long bullet/body & short Hull/Wings

negative Bfly shift ⇒ Increased curvature ⇒ long barbell/Wings & short the bullet/body.

y/c

Rem → Repo carry trade

Repo carry return = Coupon income ± Rolldown Return − Financing cost

Long futures position

$$Targeted\ Return = \%\Delta Futures\ Price - Margin\ cost$$

Video: https://youtu.be/EB9l0JwmzRA

Receive-fixed interest rate swap

swap carry

$$Targeted\ Return = (\underbrace{Swap\ rate}_{fxd} - MRR) + \%\Delta \underbrace{Swap\ mark\ to\ market}_{MTM}$$

→ based on forecast over horizon → get PV of recv s/a payout with revised swap rate = I/Y on calc ∆ adj for freq. PMT = notl × old swap rate / freq. FV = notional. Now, CPT PV. The floating rate leg of swap reset to par val → get MTM gain/loss.

to USD by mult w. Notnl

Partial Dura" / Key Rate Duration ~ apprxt non-// y/c shifts.

$$KeyRateDur_k = \frac{1}{PV} \times \frac{\Delta PV}{\Delta r_k}$$

Learning Module 6: Fixed Income Active Management: Credit Strategies

Systemic risk ≠ ≠ Systematic risk; former includes banking & g-bond risk.

$$Credit\ Spread = POD \times LGD$$

$$\text{Yield spread} = \text{YTM of credit security} - \text{YTM of benchmark bond}$$

risk in gov bond mkt ← $$G\text{-}Spread = \text{YTM of credit security} - \text{YTM of interpolated government bond}$$

risk in banking system. ← $$I\text{-}Spread = \text{YTM of credit security} - \text{Swap rate}$$

$$Asset\ swap\ spread, ASW = Bond\ coupon\ rate - Swap\ rate$$; greatest ret" when diff is highest.

Buy Recvr Swptn = rite to enter into recv fxd, pay flo swap. Sell payr Swptn ⇒ sml else bought a payr swapn — giving them the rite to pay fxd & recv floating.

Equity Idx Swap: OTC contr with c'tr party; exchng cash payout @ fixed settlmt freq. Recv Idx ret"; pay stkld (~flo) mkt's ref rate (MRR) plus mgmt fee. No trading regd by holder/long pos. → also (21)

→ ModDur ~ ½ * resetPeriod. ⇒ shortest Mdur → pfo with Max % of fltg rate bonds

FABIAN MOA

→ SpreadDur ≈ AvgMatu.

Discount margin (DM) for floating-rate notes

Calc like Eg (11)

$$PV = \frac{\left[\frac{(MRR + QM) \times FV}{m}\right]}{\left(1 + \frac{MRR + DM}{m}\right)^1} + \frac{\left[\frac{(MRR + QM) \times FV}{m}\right]}{\left(1 + \frac{MRR + DM}{m}\right)^2} + \cdots + \frac{\left[\frac{(MRR + QM) \times FV}{m}\right] + FV}{\left(1 + \frac{MRR + DM}{m}\right)^n}$$

QM = Quoted margin → Numr

DM = Discount margin → Denomr → in Eff Spdur @(36)

m = Periodicity

n = Tenor of FRN

MRR = Market reference rate (assume constant)

FV = Face value

$$Portfolio\ OAS = \sum_{i=1}^{N} \frac{MV_i}{MV} \times OAS_i$$

MV = Market value of portfolio : Bcos Wt. based on MV

$$Duration\ Times\ Spread,\ DTS = Spread\ duration \times Spread$$

↳ captures ΔVal due to Δsprd; not int rates → for term of bond ∴ identify pfo with shortest avg mat

old swap rate ∵ reset applies from subseq period.

$$\rightarrow Portfolio\ DTS = \sum_{i=1}^{N} \frac{MV_i}{MV} \times (SpreadDuration_i \times Spread_i)$$

$$Estimated\ change\ in\ portfolio\ (in\ bps) = -Portfolio\ DTS \times \%\Delta Spread$$

Impact of Yield Spreads on Portfolio Return

$$\%\Delta PV^{Spread} \approx -EffSpreadDur \times \Delta Spread + \frac{1}{2} \times EffSpreadCon \times (\Delta Spread)^2$$

$$EffSpreadDur = \frac{(PV_-) - (PV_+)}{2 \times (\Delta Spread) \times PV_0}$$

$$EffSpreadCon = \frac{(PV_-) + (PV_+) - 2(PV_0)}{(\Delta Spread)^2 \times PV_0}$$

Spread Risk : when ΔAsset yld, ΔLiab yld, ΔHedge yld differ as a result of diff mkt secn eg. IG/HY corp yld vs govt yld → @(23)

Bear steepener - steepening of y/c as rates ↑ ; Bull steepener : steepening of curve as rate ↓. ; when ST ytm fall by more than LT ytm
↳ LT ytm rise by more than ST ytm; Bull flatten : LT ytm fall by more than ST ytm.

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Excess return on credit risky bond

$$E(XR) = E(ExcessSpread) \approx (Spread_0 \times t) - (EffSpreadDur \times \Delta Spread) - (t \times POD \times LGD)$$

where:

$Spread_0$ = Initial yield spread per annum

$\Delta Spread$ = Change in spread over holding period

POD = Annualized expected probability of default

LGD = Expected loss severity

$POD \times LGD$ = expected annual credit loss

t = Holding period (in years)

Instantaneous Excess Ret'n $XR \approx Spread_0 \times t - (EffSpreadDur \times \Delta Spread)$; No adjustment for PD, LGD, from credit losses.

Video: https://youtu.be/U1C5_eNFMBA

If bond is HTM, the bond with highest spread today will have the highest E(XR) at maturity.

Floating-rate Note (FRN)

Effective rate duration ~ like @35

$$EffRateDur_{FRN} = \frac{(PV_-) - (PV_+)}{2 \times (\Delta MRR) \times PV_0}$$

Effective spread duration

DM = discount margin @35

$$EffSpreadDur_{FRN} = \frac{(PV_-) - (PV_+)}{2 \times (\Delta DM) \times PV_0}$$

Expected change in YTM based on a C% confidence interval over one month

= Daily interest rate volatility $\times \sqrt{\text{Number of trading days in a month}} \times Z\ score$

Credit Default Swap (CDS)

For a $1 notional:

$$CDS\ Price \approx 1 + [(Fixed\ Coupon - CDS\ Spread) \times EffSpreadDur_{CDS}]$$

= Upfront Prem

To pay by protect seller ← $Upfront\ premium = \{(Fixed\ coupon - CDS\ spread) \times EffSpreadDur_{CDS}\} \times Notional$

$$\%\Delta CDS\ Price \approx -\Delta(CDS\ Spread) \times EffSpreadDur_{CDS}$$

$$\Delta CDS\ Price \approx CDS\ Notional \times \%\Delta CDS\ Price$$

Recall Note: If bond spread expected to ↓ ⇒ Val ↑ ⇒ sell CDS & buy bond
" " " " ↑ ⇒ " ↓ ⇒ buy CDS ⇒ CDS net < 0 ⇒ short risk

- Buy protection (CDS) → CDS Notional < 0 (Short risk position) – if CDS curve will widen
- Sell protection → CDS Notional > 0 (Long risk position) – if CDS curve will narrow.

To calc CDS notional in L-S strat: eg. 5yr Notional = $\left(EffSpreadDur_{10} / EffSpreadDur_{5yr}\right) \times 10\text{-yr Net}$; OR 5yr Net × 5yr Dur = 10yr Net × 10yr Dur

Learning Module 7: Trade Strategy and Execution → see algos p(16)

Implementation Shortfall

$$IS = Paper\ return - Actual\ return$$

$$Paper\ return = (P_n - P_d)S$$ → current Px

$$Actual\ return = \left(\sum s_j\right)P_n - \sum s_j p_j - Fees$$ → all at current vs [actual + fees]

$$Total\ cost\ of\ paper\ portfolio = P_d \times S$$

$$IS = Execution\ cost + Opportunity\ cost + Fees$$

$$Execution\ cost = \sum s_j p_j - \left(\sum s_j\right)P_d$$ = # shares traded × (actual avg Px − Decision Px)

opp of first ≤ part in actual not

$$Opportunity\ cost = \left(S - \sum s_j\right)(P_n - P_d)$$ = (# Order − # Trad) × (Curr Px − Decision Px)

$$Expanded\ IS = Delay\ cost + Trading\ cost + Opportunity\ cost + Fees$$

Fee = # trad × fee/sh.

$$Delay\ cost = \left(\sum s_j\right)P_0 - \left(\sum s_j\right)P_d = Arrival - Decision$$

add comparison to arrival Px on top of Exec cost

Exec → add & subtr (Total × arrival Px) to get Delay + Trading

$$Trading\ cost = \sum s_j p_j - \left(\sum s_j\right)P_0 = Actual - Arrival$$

where:

P_n = Current price ; may be the closing price after order.
P_d = Decision price (Benchmark price)
P_0 = Arrival price
S = Total order of shares ($S > 0$ for buy order; $S < 0$ for sell order)
s_j = Number of shares executed in jth trade
p_j = Transaction price of jth trade
$\sum s_j$ = Total number of shares executed

Video (Buy order): https://youtu.be/xScTmNIylRQ

Video (Sell order): https://youtu.be/ssHM84hU3iw

Execution risk is imp. for trading strat; But not a factor in pre-trade blotk choice. Blotk based on decision Px or prev. close are exposed to overnight risk = prices will adjust due to new info overnight, before mrkt open ∴ eliminated by using an opening price blotk.

Traders supply liquidity by taking opp side of buy/sell pressure. If rising prices, supply liquidity by selling → maybe > mid quote.

Layering/Spoofing : traders place exposed stndg lim orders → convey liquidity or suggest secu is u/ov valued.

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VWAP algo : trade higher qty @ open & close vs smaller qty of order during midday. ∴ a U-shaped curve.

FABIAN MOA

Trade Evaluation

$$Side = \begin{cases} +1 & Buy\ order \\ -1 & Sell\ order \end{cases} \qquad \bar{P} = \text{Average execution price}$$

$$Arrival\ cost\ (bps) = Side \times \frac{\bar{P} - Arrival\ price}{Arrival\ price} \times 10{,}000$$

$$VWAP\ cost\ (bps) = Side \times \frac{\bar{P} - VWAP}{VWAP} \times 10{,}000$$

$$TWAP\ cost\ (bps) = Side \times \frac{\bar{P} - TWAP}{TWAP} \times 10{,}000$$

$$Close\ (bps) = Side \times \frac{\bar{P} - Close}{Close} \times 10{,}000$$

$$Market\ adjusted\ cost\ (bps) = Arrival\ cost\ (bps) - \beta \times Index\ cost\ (bps)$$

where:

$$Index\ cost\ (bps) = Side \times \frac{Index\ VWAP - Index\ Arrival\ Price}{Index\ Arrival\ Price} \times 10{,}000$$

⇒ If Px falls after arrival, the 2nd term < 0 ; but side of sell may again make the Numer > 0.

$$Added\ value\ (bps) = Arrival\ cost\ (bps) - Estimated\ pre\ trade\ cost\ (bps)$$

Learning Module 8: Case Study in Portfolio Management: Institutional

also @12 ← Total levered cost = Unlevered cost + Additional cost of obtaining leverage

$$\text{Additional cost of obtaining leverage for ETF} = \left(1 - \frac{1}{Leverage\ factor}\right) \times Borrowing\ cost$$

$$\text{Additional cost of obtaining leverage for Futures} = \left(1 - \frac{1}{Leverage\ factor}\right) \times Opportunity\ cost$$

$$\text{Additional cost of obtaining leverage for Swaps} = \left(1 - \frac{1}{Leverage\ factor}\right) \times Opportunity\ cost$$

Unlevered cost = Mgmt fee + commission + Px impact + Bid-ask sp + cost to roll dn(fut) + mispricing + funding cost

Portfolio Mgmt / LM05 Behavioral Biases

Cognitive errors
1) Stat, info, memory
2) faulty reasoning
3) More easily corrected
edu, info, advice

Emotional biases
1) impulse / intuition
2) feelings / emotion => spontaneous
3) less easy to correct → maybe adapted

A) Belief Perseverance biases
1) Conservatism = prior views w/ new info (under wt on past info)
2) Confo bias = look for & notice to confirm (pooling exposure to same risk // focus on info that reaffirms views only)
3) Representativeness = classify new info based on past experiences
4) Illusion of Ctrl Bias = false belief to ctrl o/comes
5) Hindsight bias = see past events as being predictable

Gambler's fallacy: recent outcome affect estim of fu prob.

B) Info-processing biases
1) Anchoring + Adjstmt Bias = incorrect psycho heuristics ≃ Conservatism // under wt on anchored value
2) Mental acctg :- treat one sum of money diffrnt from other (non fungible accts.)
3) Framing Bias - answer differently based on how ques is framed. [Narrow Framing]
4) Availability Bias - heuristic influenced by how easily outcome comes to mind.
   - recency effect
   - Base Rate Neglect - ignoring class info in favor of specific info; eg. incorporating indus data in fin forecasts to mitigate
5) Prudence Bias: over-cautious; fear of being wrong.

Emotional Biases = LOSSER
1) Loss aversion = avoid losses over gaining gains → disposition effect < hold lossy inv too long / sell gaining inv too soon
2) Overconfidence - unwarranted faith in our abilities → prediction; certainty
   - Self attribution bias = credit for gains but losses on exogenous bad luck
     - self protecting
     - self enhancing
3) Self control - fail to act thru long term goals
4) Status quo bias - do nothing vs make a change - INERTIA
5) Endowment " - value assets more when they hold rights to it
6) Regret-aversion bias - avoid pain of regret due to bad decisions

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Risk budgeting shld drive hedging strategies - not the other way round.

VaR - measures the risk of loss in a pfo over a specific time period. → ccy amt + time + prob
↳ only the min amt of ~~loss~~ expected loss; not Max.

Risk Transfer = Insurance ; Risk Shifting = Derivs

Look-Ahead bias: due to info released / updated [2.1] ex post the decision; eg. restated fin stmt

Disposition eff: behav bias when inv tend to avoid realizing losses but seek to realize gains

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